WarrantyWallet

Receipt & Warranty Management System

Design Document

Team Members

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Web Development Class | 2026

https://warranty-wallet-00hr.onrender.com

1. Project Description

WarrantyWallet is a full-stack single-page web application that helps users organize their expensive electronics and home appliances. The core problem it solves is that people frequently lose physical receipts and forget warranty expiration dates, resulting in missed claims and wasted money.

The application is split into two independent, fully functional modules:

- Purchase Wallet (Sanket Kothari) — A digital receipt box that tracks what users own, what they paid, and where they bought it. Users can upload receipt images and filter their purchase history.
- Warranty Vault (Jinam Shah) — A support library that stores warranty terms, PDF manuals, support contacts, and calculates real-time countdown timers for warranty expiration dates. Automated email reminders alert users before warranties expire.

Both modules are connected to the same MongoDB Atlas database and Cloudinary file storage, but operate independently — adding a purchase does not require a warranty entry and vice versa.

2. Technology Stack

Layer	Technology	Purpose
Frontend	Vanilla JavaScript ES6	Client-side rendering, DOM manipulation
Styling	CSS3 (Modular)	wallet.css, support.css, style.css, settings.css
Backend	Node.js + Express.js	REST API server
Database	MongoDB (Native Driver)	Two collections: purchases, support_docs
File Storage	Cloudinary	Receipt images and PDF manual storage
Email	Nodemailer (Gmail)	Automated warranty reminder emails
Scheduler	node-cron	Daily warranty check at 9:00 AM
Linting	ESLint v9	Code quality enforcement
Formatting	Prettier	Consistent code formatting
Deployment	Render.com	Public cloud hosting

3. User Personas

Persona 1 — Jason (Tech Owner)

Attribute	Details
Age	28
Occupation	Software Engineer
Tech Comfort	High
Problem	Owns 15+ gadgets, keeps receipts in a drawer, forgets warranty dates
Goal	A central place to upload receipts and track warranty status for all devices
Quote	I just found out my laptop warranty expired last month. I could have claimed a free repair.

Persona 2 — Sarah (Frustrated User)

Attribute	Details
Age	35
Occupation	Homemaker
Tech Comfort	Medium
Problem	Blender broke, can't find the manual or customer support number
Goal	Quick access to support contacts and manuals when appliances break
Quote	I spent 2 hours searching for the warranty card. It was in a kitchen drawer.

Persona 3 — Mike (Busy Parent)

Attribute	Details
Age	42
Occupation	Account Manager
Tech Comfort	Low-Medium
Problem	Buys electronics monthly for family, no system to track warranties
Goal	Know instantly if a product is still under warranty before calling for repairs
Quote	Is the TV still under warranty? I have no idea — I threw away the box.

4. User Stories

Purchase Wallet — Sanket Kothari

#	User Story	Priority
1	As a user, I want to add a purchase (item, store, price, date, category) so I have a complete purchase history.	High
2	As a user, I want to upload a receipt photo so I have digital proof of purchase for returns or insurance.	High
3	As a user, I want to see all my purchases in a sortable table so I can review my spending history.	High
4	As a user, I want to filter purchases by category (Electronics, Home, Appliances) so I can organize my inventory.	Medium
5	As a user, I want to search purchases by item name or store so I find specific records quickly.	Medium
6	As a user, I want to see total spending stats (total spent, count, by category) so I understand my spending patterns.	Medium
7	As a user, I want to edit purchases so I can correct mistakes or update information.	High
8	As a user, I want to delete purchases for items I no longer own so my inventory stays current.	High

Warranty Vault — Jinam Shah

#	User Story	Priority
1	As a user, I want to add a warranty document (product, brand, expiry date) so I track coverage for all my items.	High
2	As a user, I want to upload a PDF manual so I can access product documentation without keeping paper copies.	High
3	As a user, I want to see a live warranty countdown (days remaining) so I know exactly how long coverage lasts.	High
4	As a user, I want status badges (Active, Expiring Soon, Expired) so I can see warranty health at a glance.	High
5	As a user, I want to store support contact info (phone, email, website) so I know exactly who to call when things break.	Medium
6	As a user, I want to search the library by brand so I quickly find warranty docs for specific manufacturers.	Medium
7	As a user, I want to receive automated daily email reminders before warranties expire so I never miss a claim deadline.	High
8	As a user, I want to set my notification email from a settings panel so I control where reminders are sent.	Medium

9	As a user, I want to trigger a reminder email manually so I can test that the system is working.	Low
10	As a user, I want to edit and delete warranty documents so my library stays accurate.	High

5. Application Architecture

WarrantyWallet follows a Single Page Application (SPA) architecture where the frontend is built entirely with Vanilla JavaScript and communicates with the backend through JSON REST APIs.

Layer	Files	Responsibility
Frontend HTML	public/index.html	Single HTML file, all views rendered dynamically
Frontend CSS	wallet.css, support.css, style.css, settings.css	Modular CSS per component
Frontend JS	wallet.js, support.js, app.js, settings.js	Vanilla JS modules, client-side rendering
Backend Routes	walletRoutes.js, supportRoutes.js, emailRoutes.js	Express REST API endpoints
Database	db/connect.js	MongoDB native driver connection
File Storage	db/cloudinary.js	Cloudinary upload configuration
Email Service	services/reminderJob.js	Nodemailer + cron scheduler
Entry Point	server.js	Express server initialization

MongoDB Collections

Collection	Owner	Key Fields
purchases	Sanket Kothari	itemName, storeName, price, purchaseDate, category, receiptFile
support_docs	Jinam Shah	productName, brand, warrantyExpiry, daysLeft, status, manualFile, supportPhone
settings	Shared	key: notificationEmail, value: user@email.com

API Endpoints

Method	Endpoint	Description	Owner
GET	/api/purchases	Get all purchases	Sanket
GET	/api/purchases/stats/summary	Get spending stats	Sanket
POST	/api/purchases	Create purchase	Sanket
PUT	/api/purchases/:id	Update purchase	Sanket
DELETE	/api/purchases/:id	Delete purchase	Sanket
GET	/api/support	Get all warranty docs	Jinam
GET	/api/support/stats/summary	Get warranty stats	Jinam

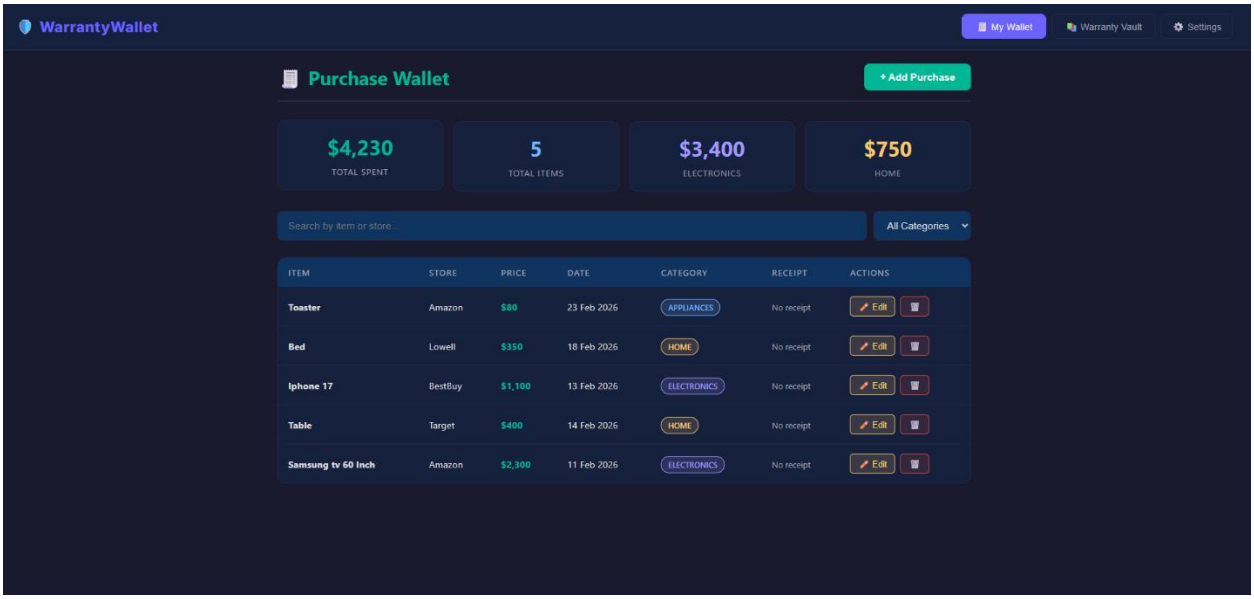
GET	/api/support/search/:brand	Search by brand	Jinam
POST	/api/support	Create warranty doc	Jinam
PUT	/api/support/:id	Update warranty doc	Jinam
DELETE	/api/support/:id	Delete warranty doc	Jinam
GET	/api/email	Get notification email	Sanket
POST	/api/email	Save notification email	Sanket
POST	/api/email/test	Trigger reminder email	Sanket

6. Design Mockups & Screenshots

The following section shows the UI design of WarrantyWallet. Please add screenshots from the live application at https://warranty-wallet-00hr.onrender.com

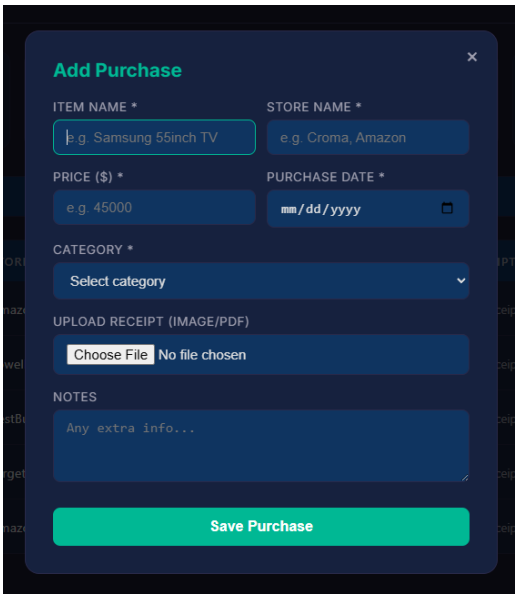
6.1 Purchase Wallet Dashboard

The Purchase Wallet displays a stats bar at the top showing total spending, item count, and spending by category. Below is a searchable, filterable table of all purchases with receipt thumbnails.



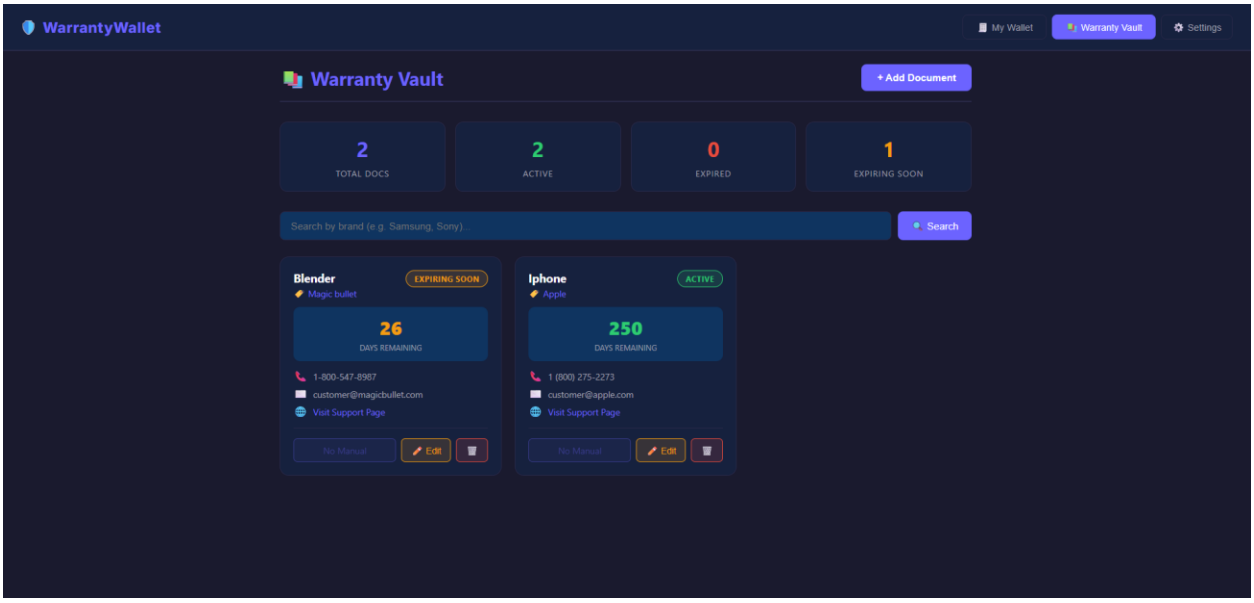
6.2 Add Purchase Modal

The Add Purchase modal contains a form with fields for item name, store, price, purchase date, category, receipt upload, and notes. All required fields are marked with an asterisk.



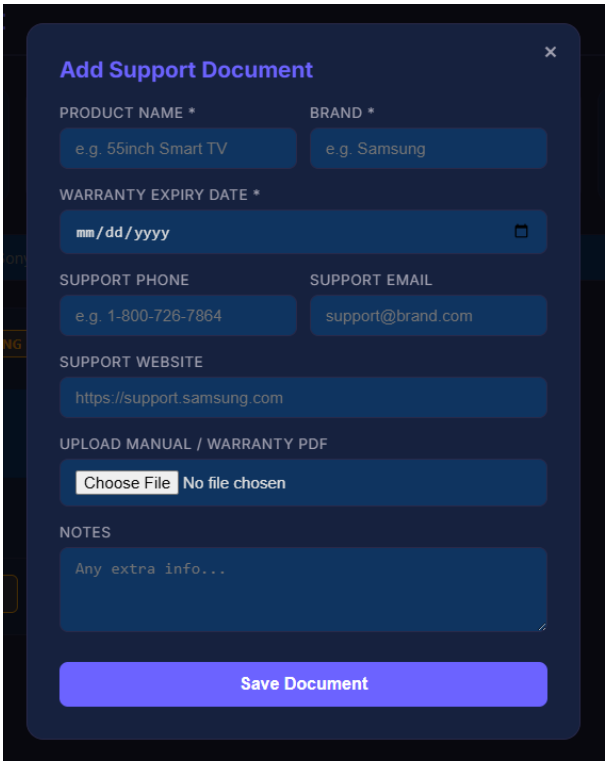
6.3 Warranty Vault Dashboard

The Warranty Vault shows a stats bar with total docs, active, expired, and expiring soon counts. Cards display product name, brand, status badge, live countdown timer, and support contact information.



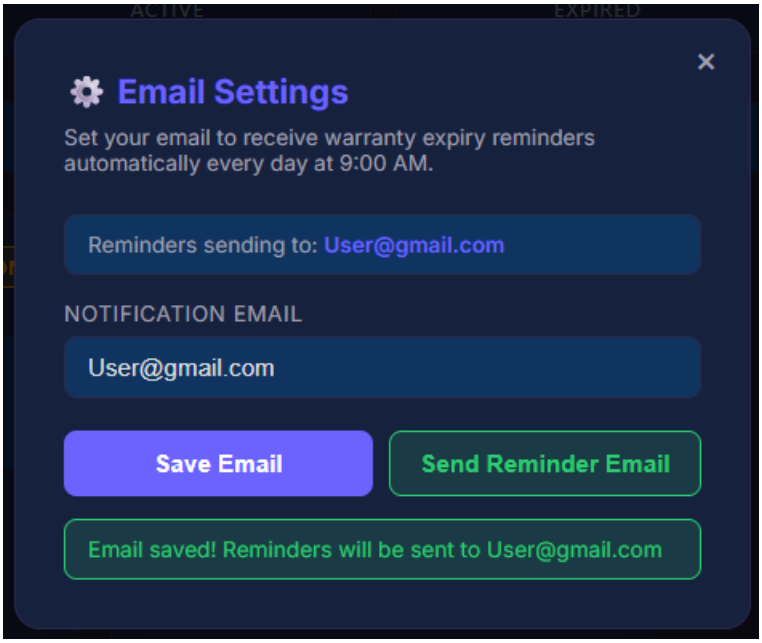
6.4 Add Warranty Document Modal

The Add Document modal allows users to enter product name, brand, warranty expiry date, support phone, email, website, and upload a PDF manual.



6.5 Email Settings Modal

The settings modal allows users to set their notification email address and trigger a manual warranty reminder email.



7. Work Distribution

Member	Module	Files Owned	Responsibilities
Sanket Kothari	Purchase Wallet	walletRoutes.js, wallet.js, wallet.css, emailRoutes.js	Purchases CRUD, receipt upload, spending stats, search/filter, email API
Jinam Shah	Warranty Vault	supportRoutes.js, support.js, support.css, style.css, app.js, settings.js, settings.css, reminderJob.js	Warranty CRUD, manual upload, countdown timer, email reminders, settings UI, app router

8. Deployment

Item	Details
Platform	Render.com (Free Tier)
Live URL	https://warranty-wallet-00hr.onrender.com
GitHub Repo	https://github.com/Reachout-git-sk/Warranty-Wallet
Database	MongoDB Atlas — Free M0 Cluster (AWS us-east-1)
File Storage	Cloudinary (Free Tier)
Auto Deploy	Enabled — pushes to main branch trigger automatic redeploy
Build Command	npm install
Start Command	node server.js

9. Setup Instructions

To run this project locally:

1. Clone the repository: git clone https://github.com/Reachout-git-sk/Warranty-Wallet.git
2. Install dependencies: npm install
3. Create .env file with: MONGO_URI, CLOUDINARY_CLOUD_NAME, CLOUDINARY_API_KEY, CLOUDINARY_API_SECRET, EMAIL_USER, EMAIL_PASS
4. Start the server: node server.js
5. Open browser at: http://localhost:3000

10. License

This project is licensed under the MIT License. See the LICENSE file in the repository for full details.

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